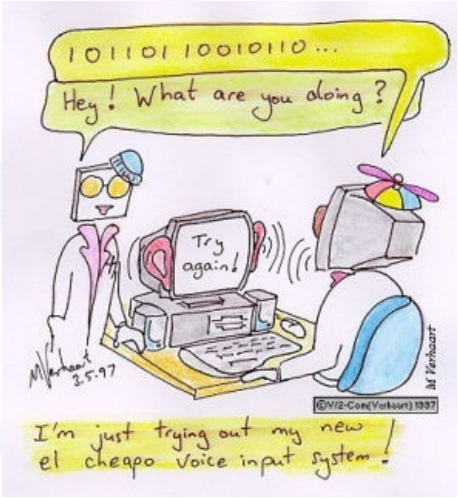
Signals and Systems

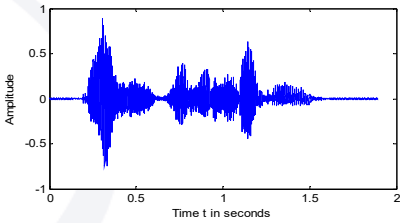
Module 8 Sampling and Reconstruction



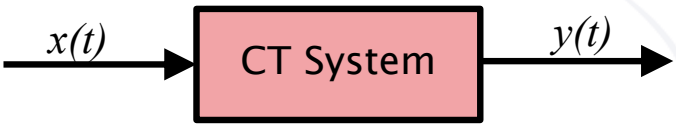
Overview

- This module will cover the following topics:
  - Sampling of Continuous Signals
  - Shannon-Nyquist Theorem and Nyquist rate
  - Aliasing
  - Reconstruction of sampled signals
  - Practical sampling and reconstruction considerations

Continuous vs Discrete time



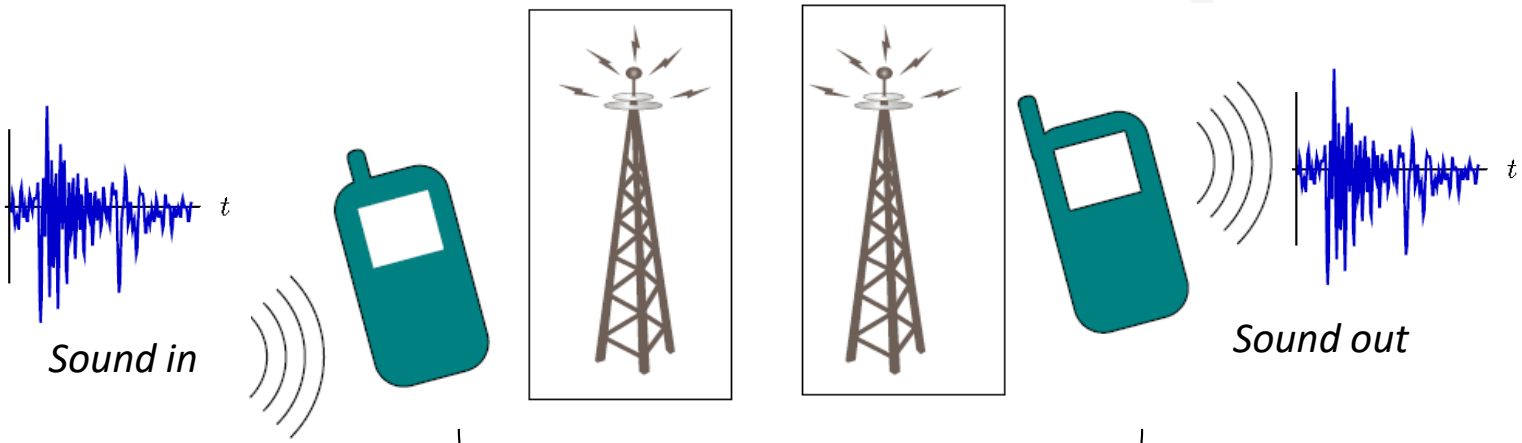
- Most physical signals are Continuous Time signals
- CT systems used to modify these signals



- Increasingly signals are converted to digital (discrete time) signals for processing then converted back

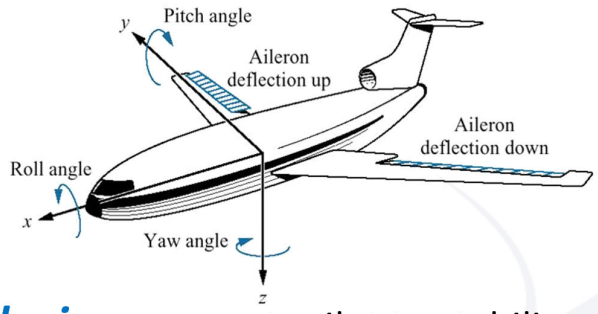


Example – Mobile phone system

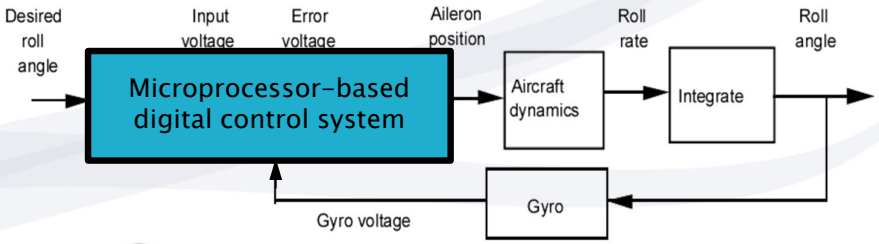


Digital / digitally encoded signals

Example – Aircraft



- ***Control system design:*** e.g. autopilot to stabilise the roll



Advantages of digital systems

- Signal manipulations more easily performed using computers than analogue components
- Easier to change systems and responses (even in real-time)
- System responses can be reproduced more consistently
- More compact implementations
- Signals can be retransmitted without degradation (loss/noise)
- Easier to store signals for later use

…1001001110101011…



Digital signal transmission

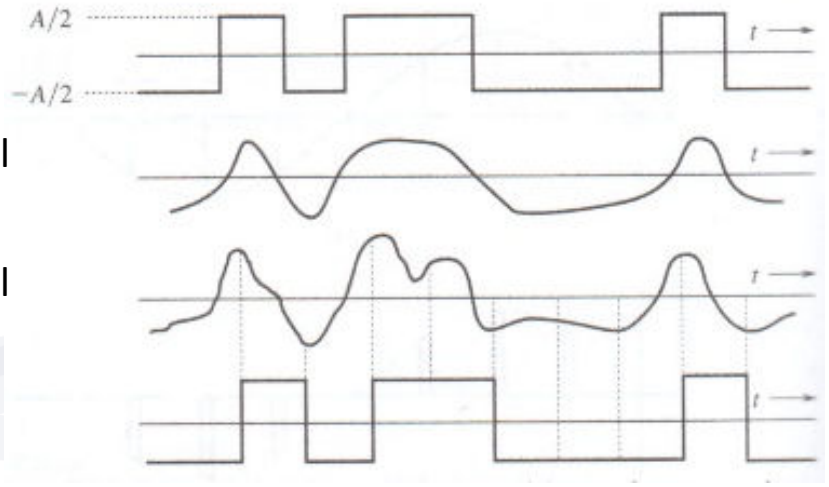
- Signal can be regenerated without degradation

a) at the transmitter

b) Received distorted signal (without noise)

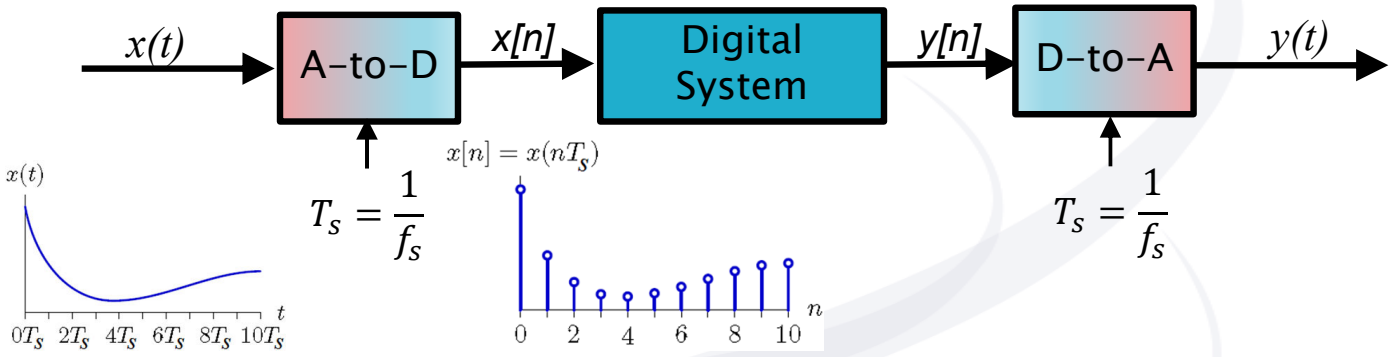
c) Received distorted signal (with noise)

d) Regenerated signal at receiver



Lathi 8.3 p792

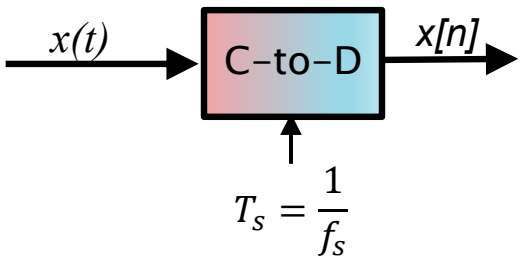
Continuous / discrete conversion



Sampling **Reconstruction**

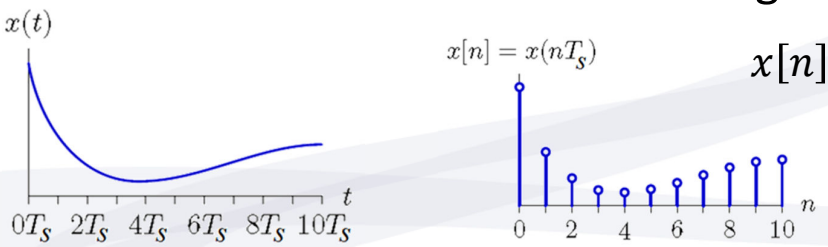
Ideal sampling

Continuous to Discrete



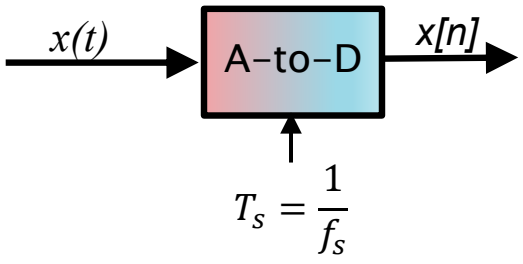
- Discrete-time signal is snapshot of continuous time signal at regularly spaced intervals
- Signal has continuous range

$$x[n] = x(nT_s), \quad n = \cdots, -1, 0, 1, \ldots.$$

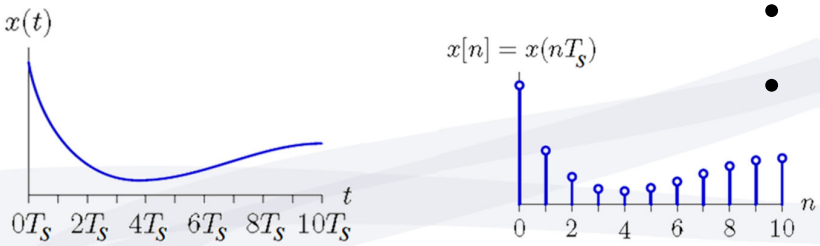


Practical sampling

Analogue to Digital

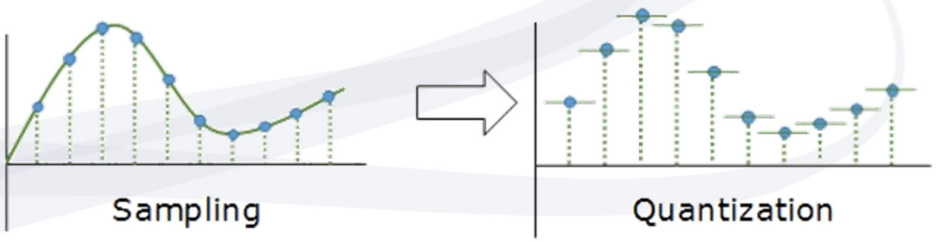


- Digital signal is approximation of ideal discrete time signal
- Signal may be degraded by real-world problems such as:
  - Word-length quantisation error
  - Sampling time jitter

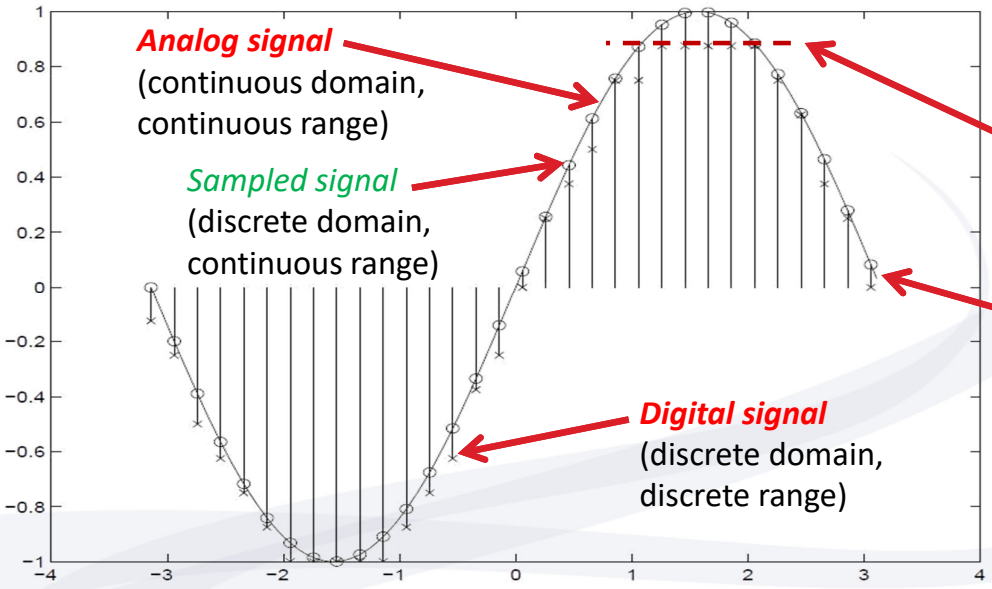


Quantisation error

- A finite number of bits used to represent digital signals
- ∴ Signal magnitude can only take on discrete set of values
- N bits = 2^N levels
- For linear scheme, level 'step' = $\frac{full\ range}{2^N}$

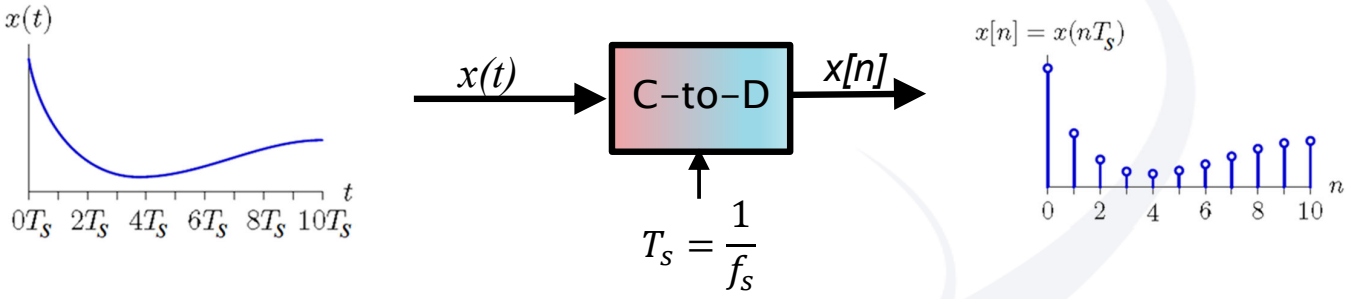


Quantisation error



- If signal too large, signal gets clipped (distortion)
- For small signals, relative error is large

Sampling



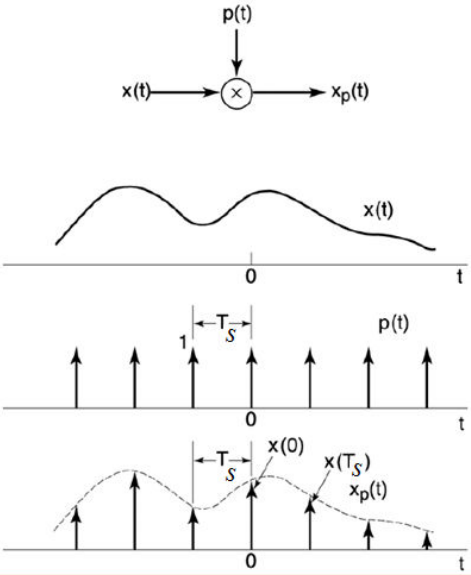
- Sampling is just another system 'block' that transforms the input signal
- Therefore has a frequency response / transfer function

Sampling

- Sampling is equivalent to multiplying by impulse train with period T_s
- But multiplication in time domain = convolution in frequency domain
  - (refer Module 6)

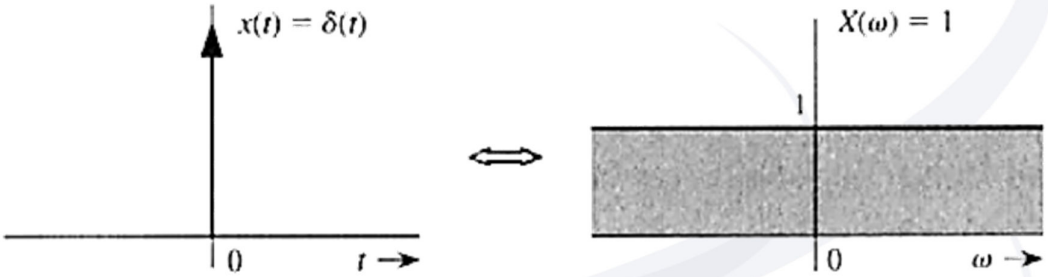
$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

$$x_p(t) = x(t)p(t) = \sum_{n=-\infty}^{\infty} x(t)\delta(t - nT_s) = \sum_{n=-\infty}^{\infty} x(nT_s)\delta(t - nT_s)$$



Recap: Fourier transform of Unit Impulse

$$\mathcal{F}[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1$$

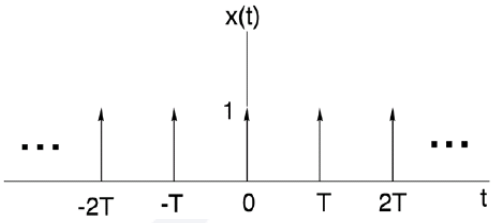


- The spectrum of the unit impulse contains frequency components at all frequencies
  - The narrower the time-pulse, the wider the spectrum

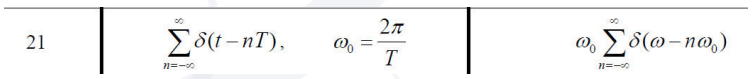
Sampling function spectrum

- Sampling function

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

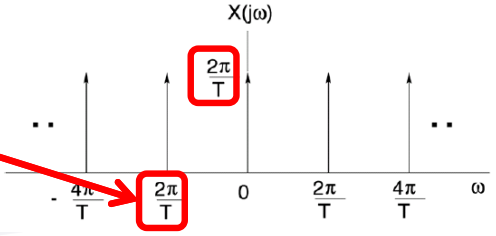


- Spectrum of the sampling function
  - (using Pair 21 on formula sheet)



$$X(j\omega) = \frac{2\pi}{T_s} \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s)$$

$$\omega_s = \frac{2\pi}{T_s}$$



  - Impulse every ω_s, scaled by factor $\frac{2\pi}{T_s}$

Sampled spectrum

- Sampling = multiplying by impulse train with period T_s
- Multiplication in time domain = convolution in frequency domain
- From formula sheet *(frequency convolution operation)* :

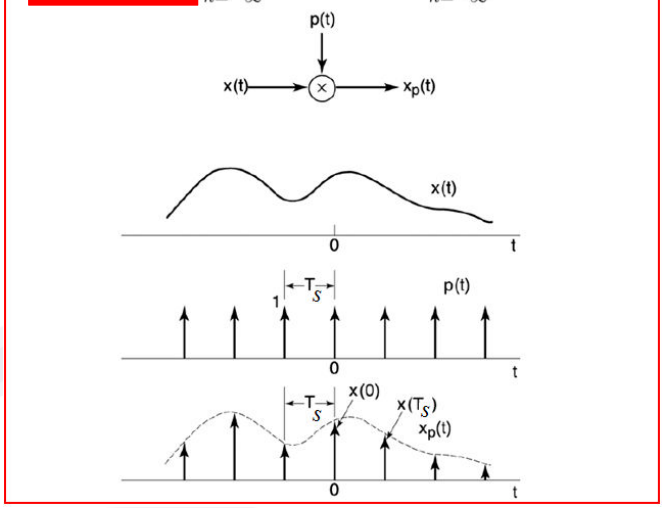
$x_1(t)x_2(t)$	$\frac{1}{2\pi}X_1(\omega) * X_2(\omega)$

$$x_p(t) = x(t)p(t)$$

$$X_p(j\omega) = \frac{1}{2\pi}X(j\omega) * P(j\omega)$$

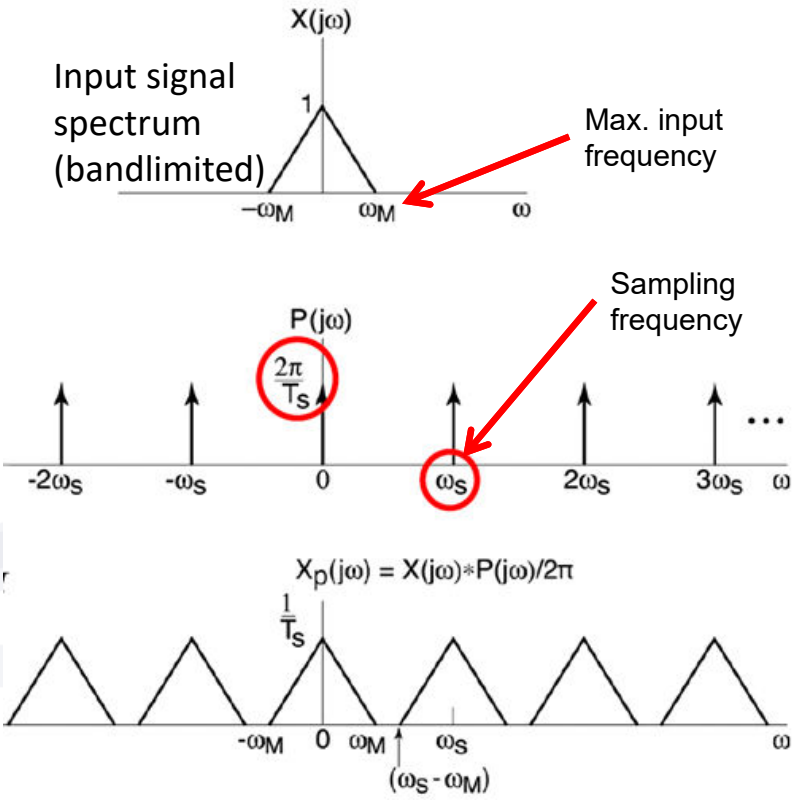
$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

$$x_p(t) = x(t)p(t) = \sum_{n=-\infty}^{\infty} x(t)\delta(t - nT_s) = \sum_{n=-\infty}^{\infty} x(nT_s)\delta(t - nT_s)$$

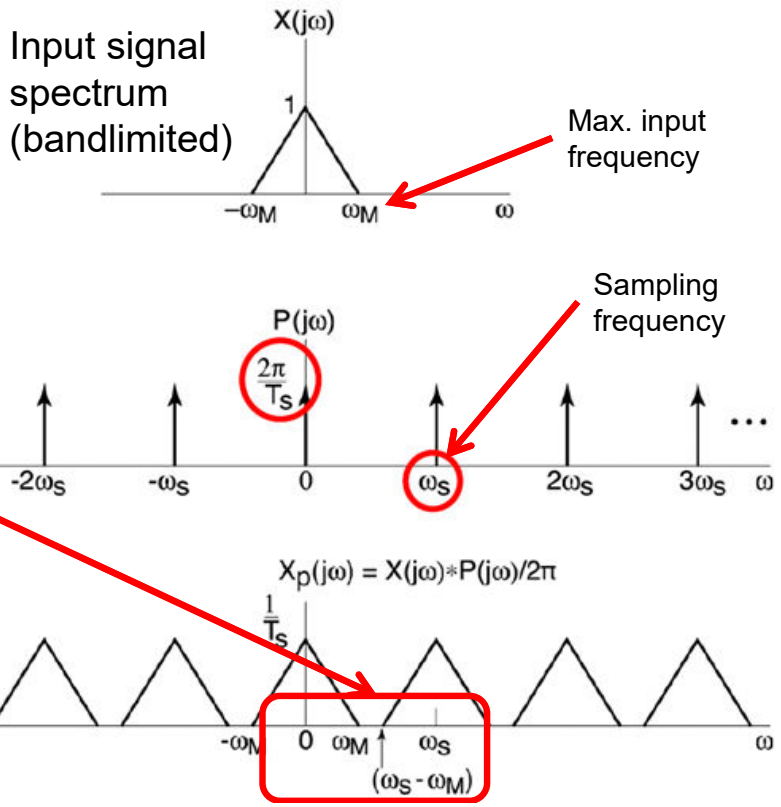


Sampled spectra

- In frequency domain, sampling is convolution of signal and impulse train
- Input spectrum replicated every ω_s
- Magnitude scaled by $\frac{1}{T_s}$



Sampled spectra



- For the replicated spectra not to overlap, need:

$$\omega_s - \omega_m > \omega_m$$

$$\therefore \omega_s > 2\omega_m$$

Shannon–Nyquist Sampling Theorem

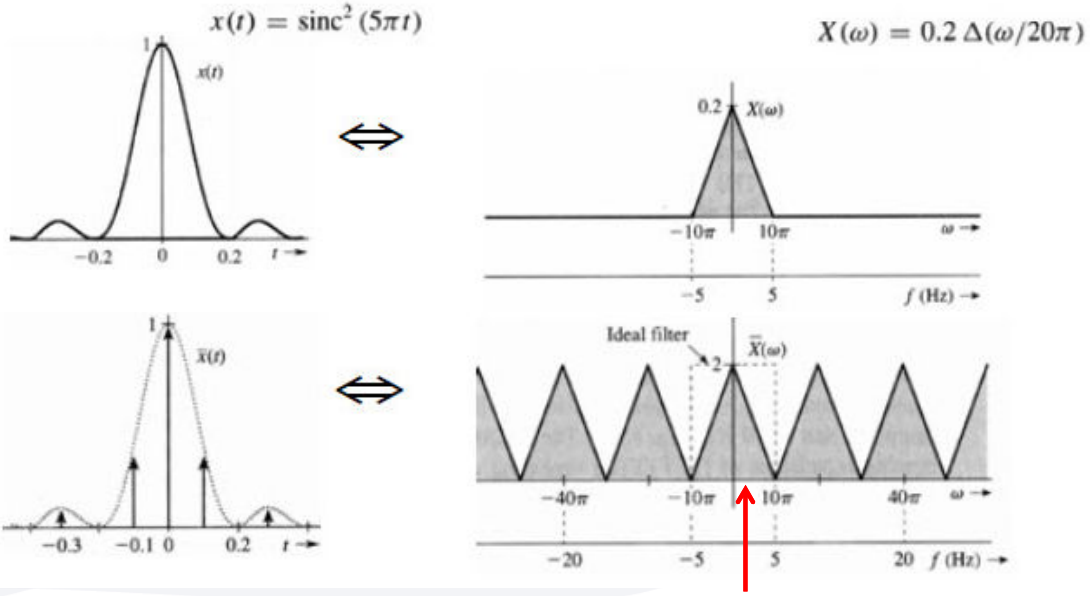
- If a signal *x(t)* has highest frequency component = *B* hertz
- Perfect reconstruction is possible if sampling frequency $f_s > 2B$

Nyquist Rate: $f_s = 2B$

Nyquist interval: $T_s = \frac{1}{2B}$

Sampling at Nyquist rate

Consider a signal bandlimited to 5Hz



Sampling at Nyquist rate (10 Hz)

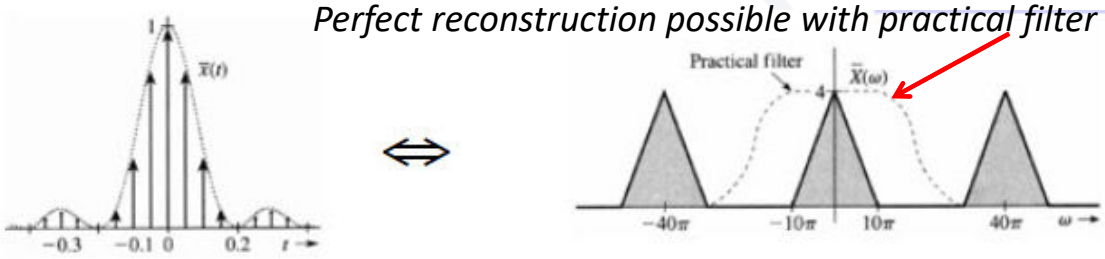
Perfect reconstruction possible with ideal filter

Lathi 8.1 p774

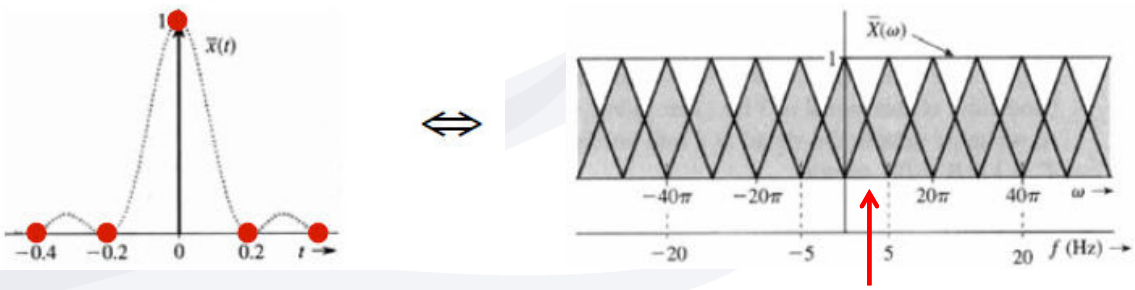
Oversampling and undersampling

Perfect reconstruction possible with practical filter

Sampling higher than Nyquist rate (20 Hz)



Sampling below Nyquist rate (5 Hz)



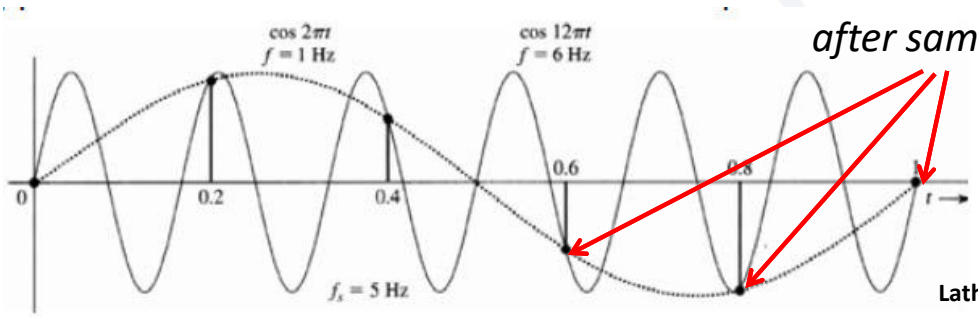
ALIASING

Signal corrupted

Lathi 8.1 p774

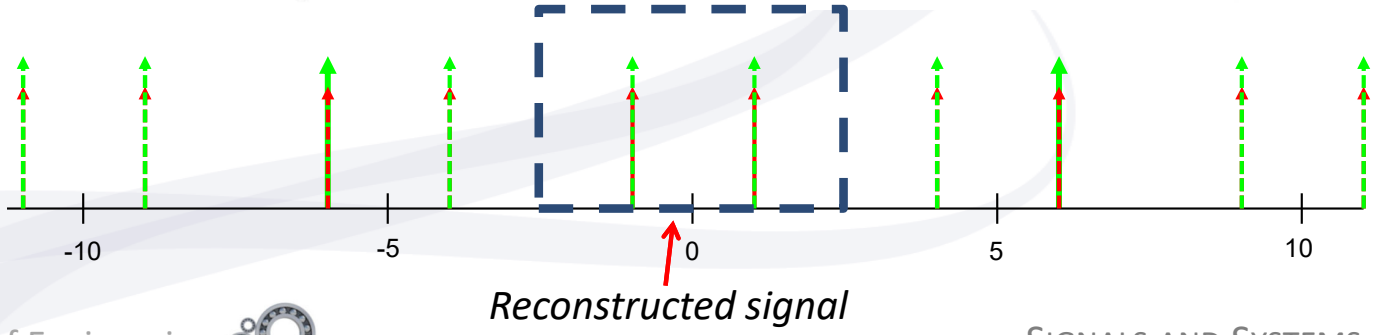
Spectral folding effect of Aliasing

1 Hz sinewave and
6 Hz sinewave
sampled at 5 Hz



Results the same after sampling

Lathi 8.2 p788



Reconstructed signal

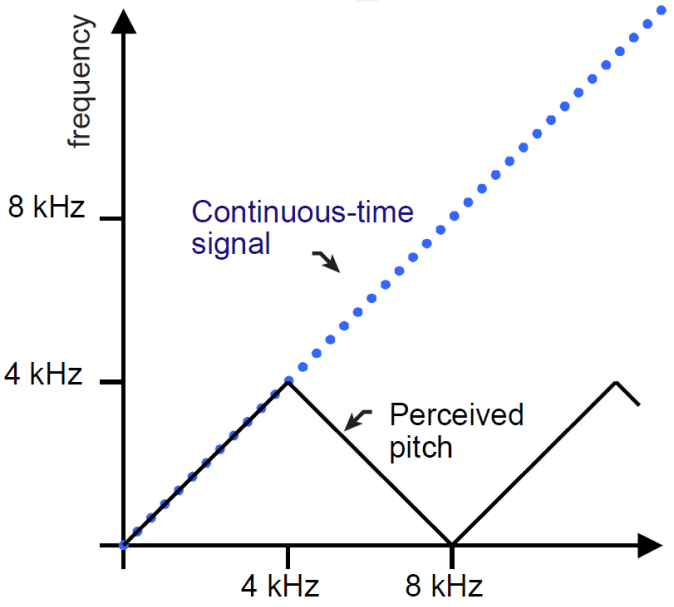
Spectral folding effects

- Consider a sinusoid of frequency f sampled at f_s samples/sec,
- Sampled version would appear to be sample of continuous-time sinusoid of frequency $|f_a|$ in the band 0 to $f_s/2$.

$$|f_a| = |f - mf_s| \qquad \text{where } |f_a| \leq \frac{f_s}{2},\ m \text{ an integer}$$

Spectral folding effects

- Example: A signal of continuously increasing frequency is sampled at 8 kHz
- The reconstructed signal will never exceed 4 kHz
- After 4 kHz, apparent frequency will go down to 0, then back up again.



Wagon wheel effect

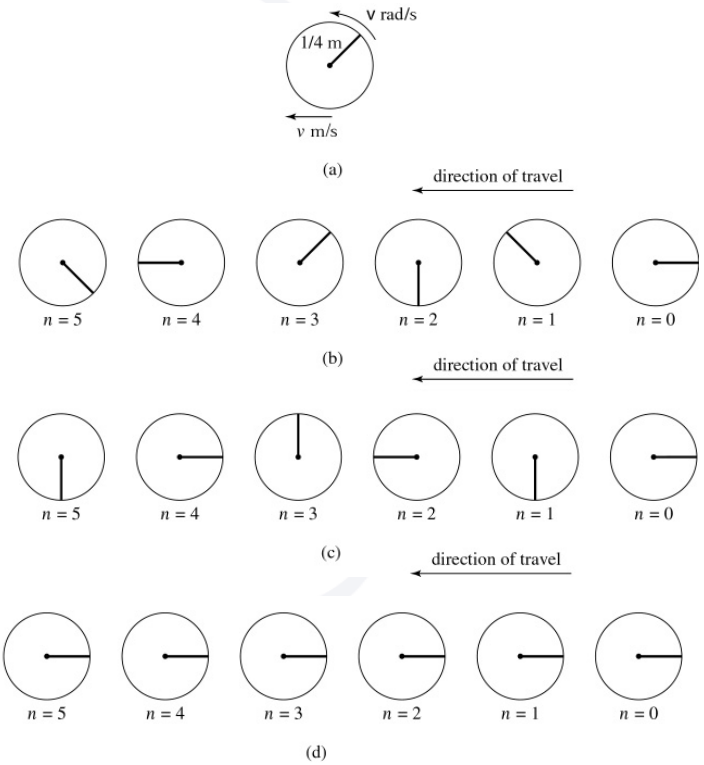
- Aliasing in a movie

(a) Wheel moving right to left at v ms^{-1}

(b) Sequence of frames when wheel rotates less than half a turn per frame

(c) Sequence of frames when wheel rotates more than half a turn per frame

(d) Sequence of frames when wheel rotates one turn per frame



Wagon wheel effect

- Aliasing in a movie
- Example:

  http://www.michaelbach.de/ot/mot_wagonWheel/index.html

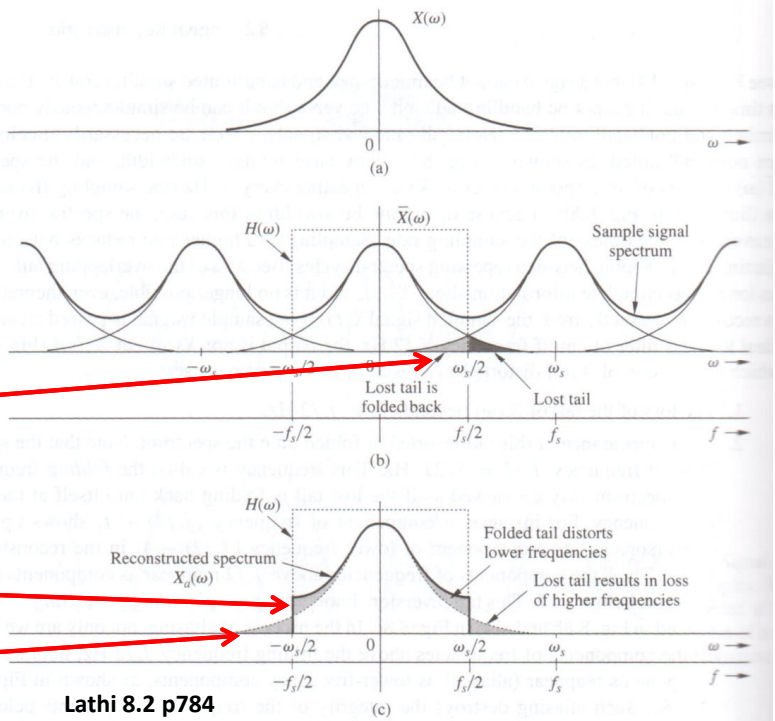
Anti-aliasing filter

Consider a signal with bandwidth greater than $f_s/2$

After sampling:
Aliasing occurs with 'tail' (frequencies > $f_s/2$)

After reconstruction:

a) Lower frequency components distorted due to 'folded tail'
b) loss of frequency components higher than $f_s/2$

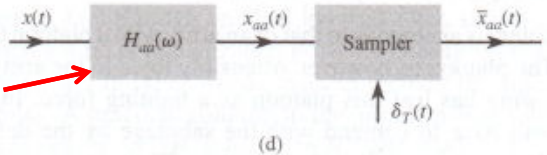


Lathi 8.2 p784

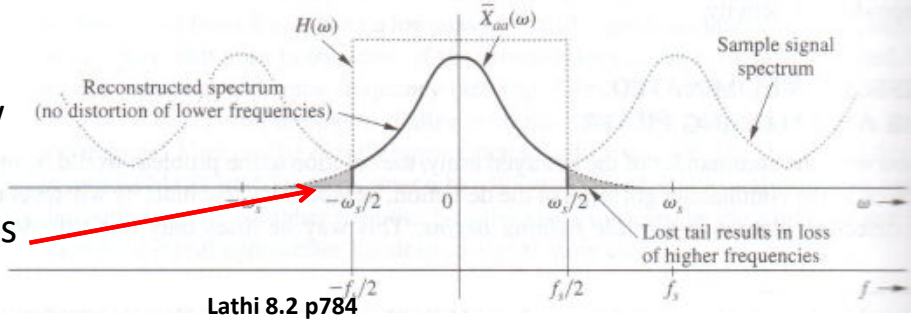
Anti-aliasing filter

Solution: Apply a low-pass filter that limits bandwidth to $f_s/2$ before sampling

Anti-aliasing filter



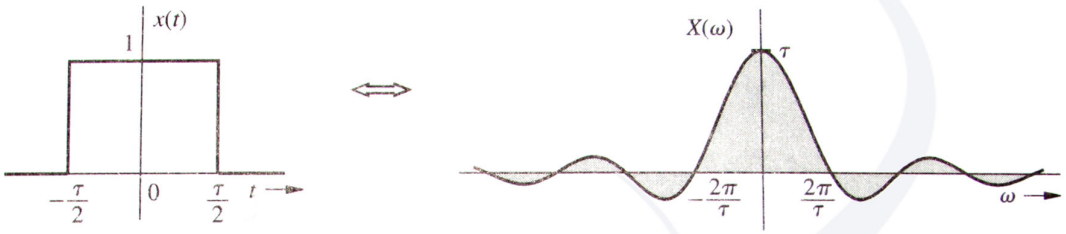
(d)

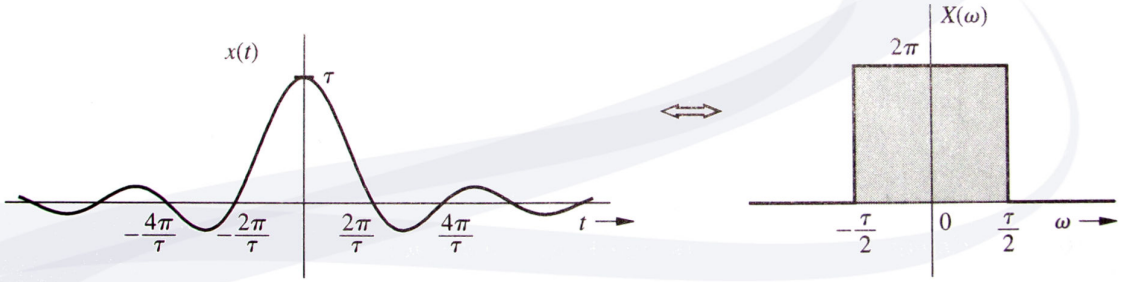


- No more distortion of frequency components below $f_s/2$
- Loss of frequency components higher than $f_s/2$ still occurs

Lathi 8.2 p784

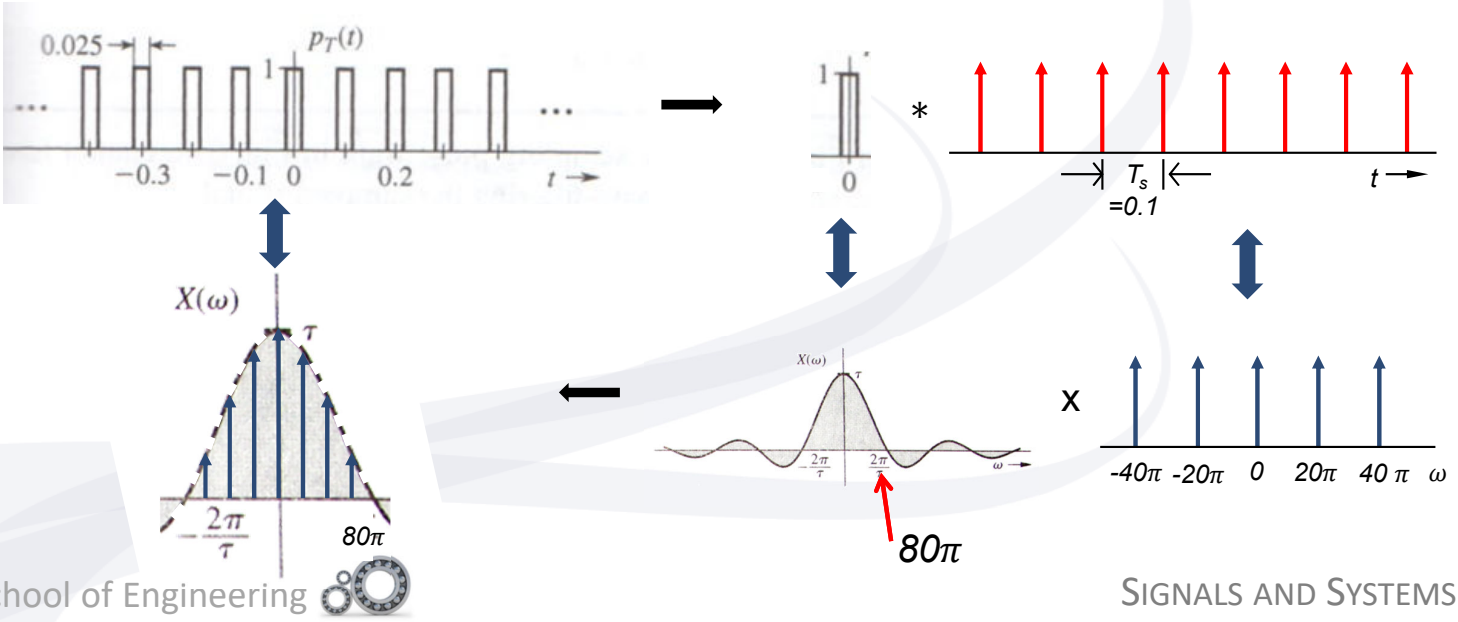
Recap: Time-Frequency Duality





Finite pulse spectra

- Consider a pulse train $p_T(t)$ with frequency 10 Hz and pulse width τ = 0.025 sec

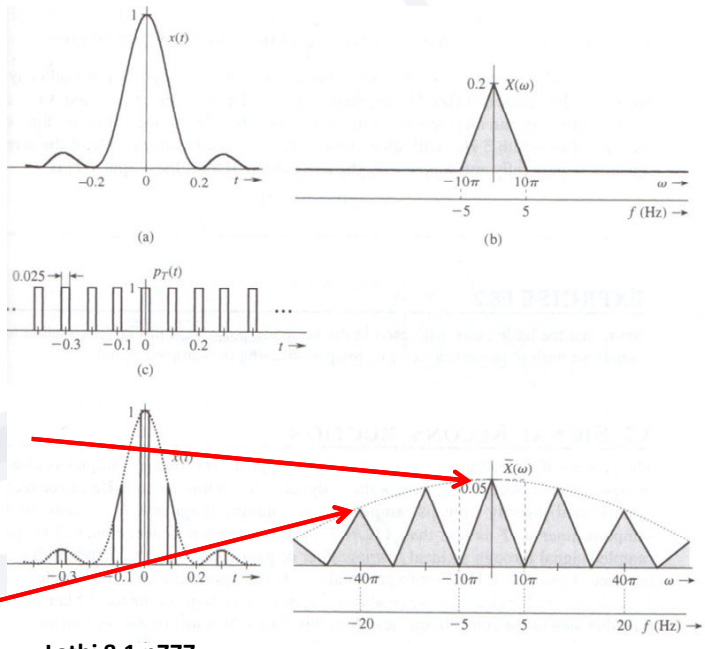


Practical sampling

- Impulse train not a very practical sampling signal

- Use pulses of finite width
- 'Non-flat' spectral envelope

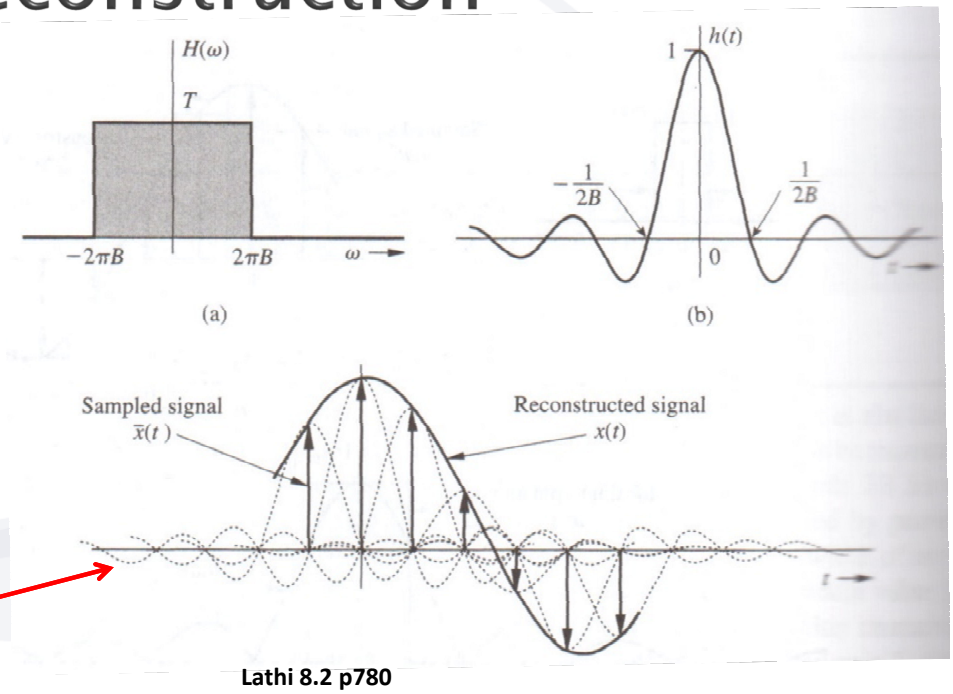
- Slight roll-off in baseband frequency response
  - narrower pulse, less roll-off
- Reduces gain of repeated spectra



Lathi 8.1 p777

Ideal Signal Reconstruction

- Pass sampled signal through ideal lowpass filter, bandwidth B
- B is bandwidth of initial signal, $B \leq \frac{f_s}{2}$

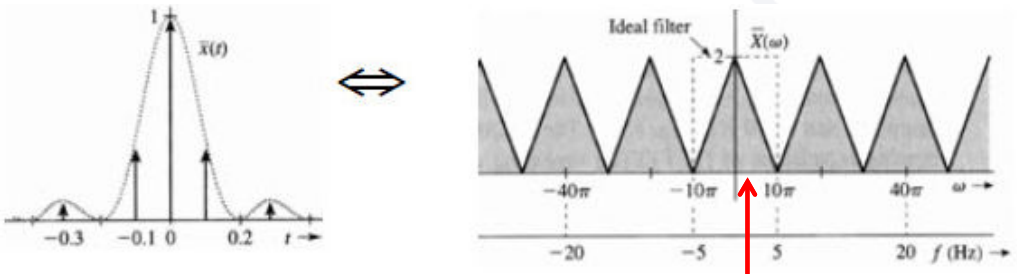


That's why Sinc function also known as **interpolation** function

Lathi 8.2 p780

Practical signal reconstruction

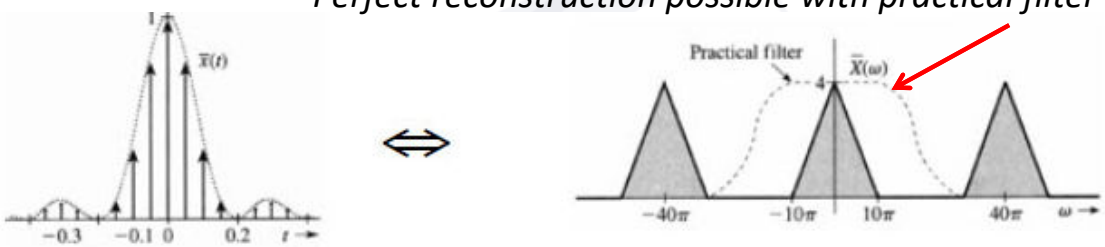
Sampling at Nyquist rate



Perfect reconstruction possible with ideal filter

Perfect reconstruction possible with practical filter

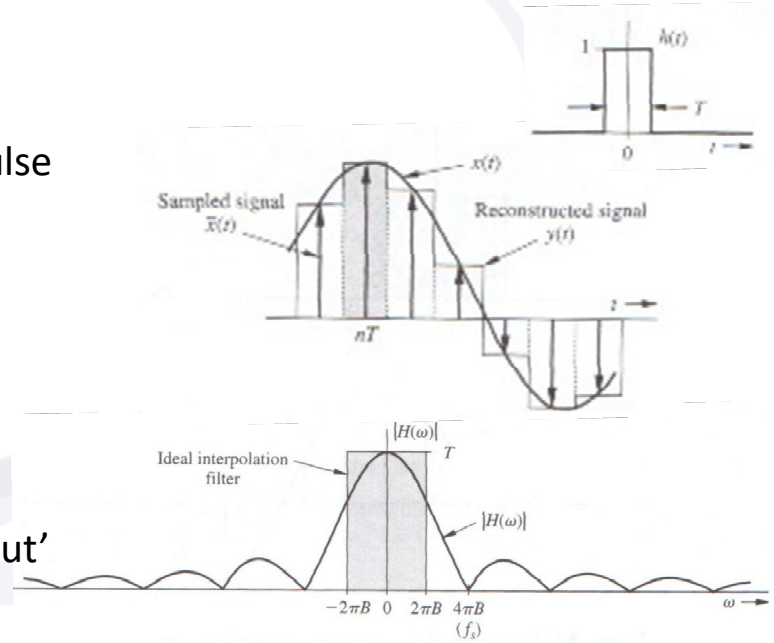
Sampling higher than Nyquist rate



Lathi 8.1 p771

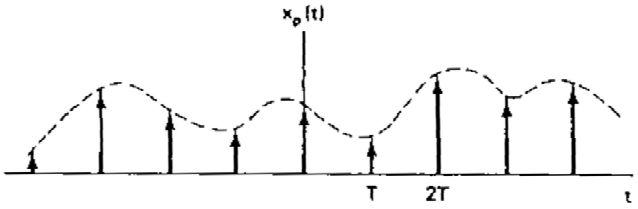
Reconstruction – zero-order hold

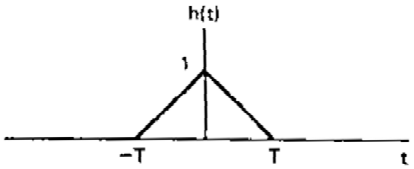
- D-A converter
  - Simple interpolator
  - Each sample converted to a pulse of width T_s
  - Performs job of signal reconstruction
  - Called **zero-order hold** circuit
  - Can apply post-reconstruction low-pass filtering to 'smooth out' reconstructed signal

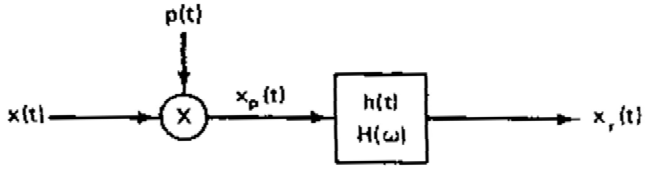


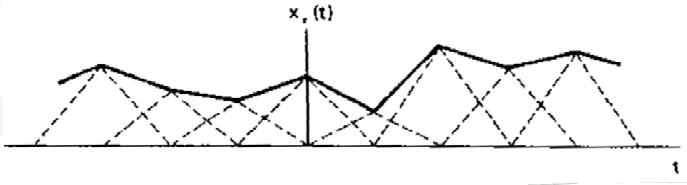
Reconstruction – first-order hold

- Alternative
  - Linear interpolator
  - Triangular impulse response
  - Called **first-order hold** circuit









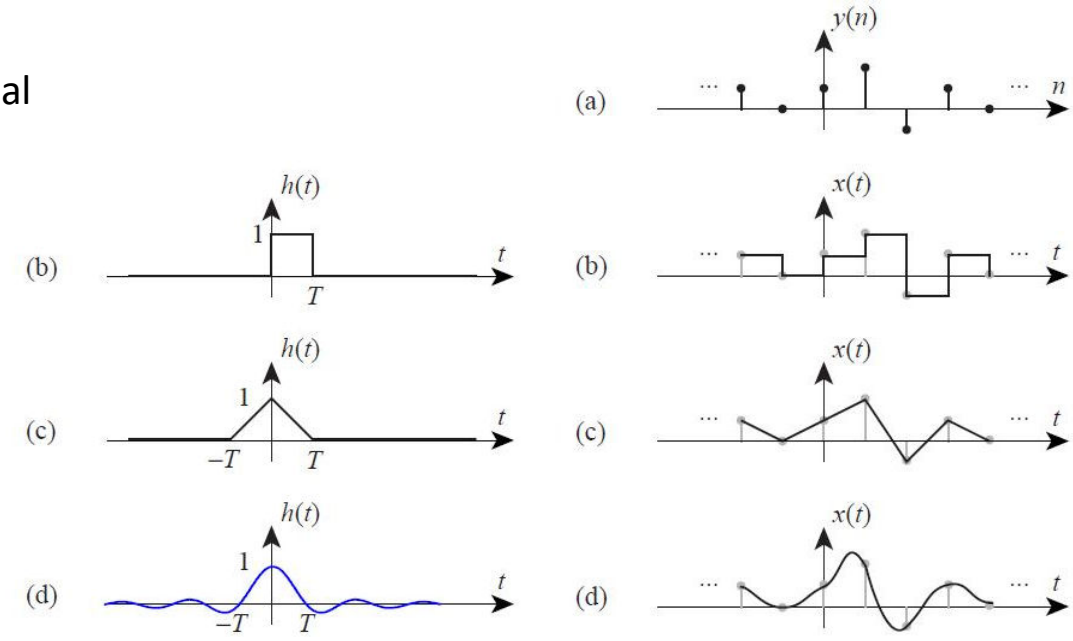
Reconstruction – comparison

Discrete-time signal

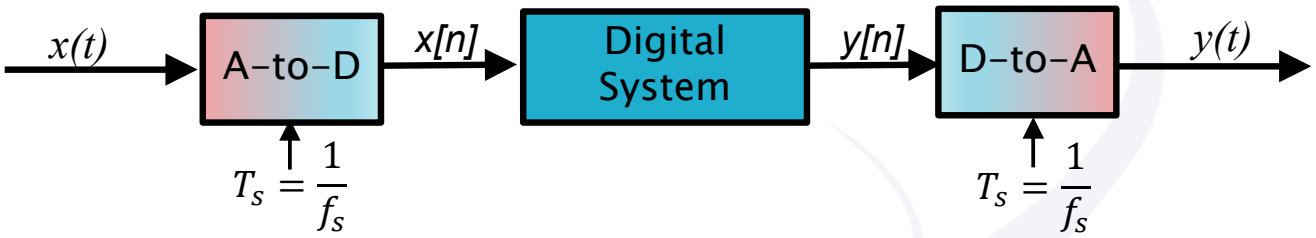
Zero-order hold

First-order hold

Ideal (sinc) interpolation



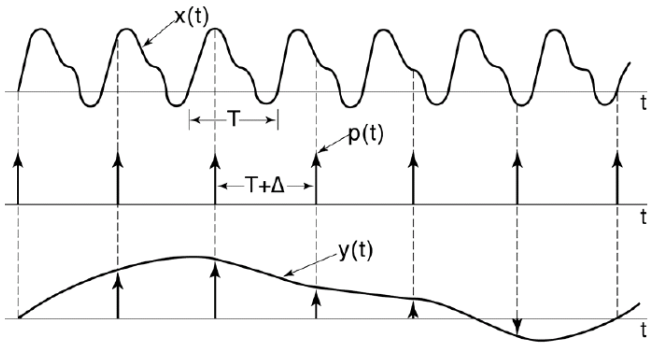
Practical considerations

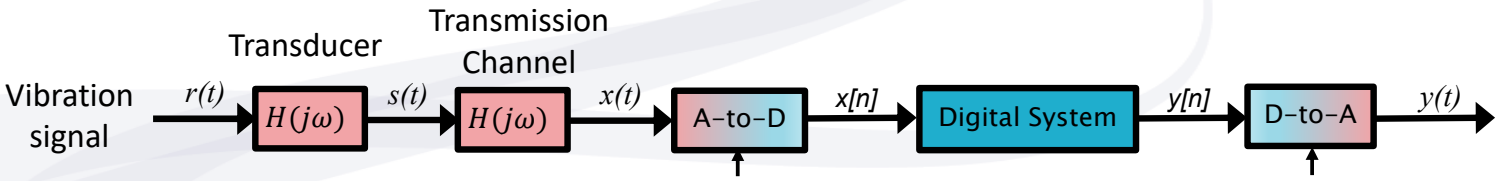


- Sampling and reconstruction are ‘sub-systems’
- Their impulse/frequency ‘response’ affects signals
- Should also consider all other components in the system
- Need to be aware of their effects and take into consideration

Example – Vibration sensor

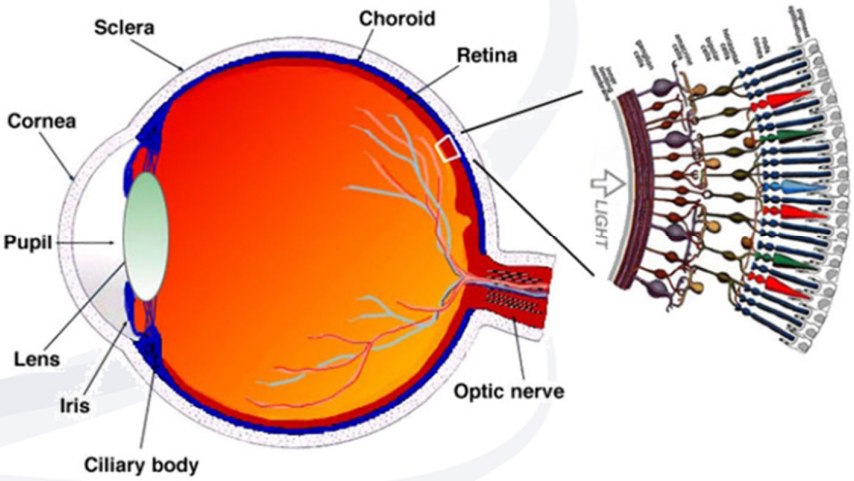
- Sampling rate too low
- Miss out key elements of signal
- Need to also consider frequency response of other components
  - E.g. transducer, channel characteristics





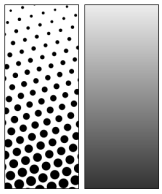
Example – human eye

- Every image that we see is sampled by the retina
  - Contains ≈ 100 million rods and 6 million cones
    - average spacing ≈ 3mm
- Act as discrete sensors
- Encoded signals sent to brain



Example – human eye

- Photographs in newsprint: *halftone* images
- Each point is black or white and the average conveys brightness
- Uses eye 'system response'



Left: Halftone dots

Right: How the human eye would see this from a sufficient distance.



- Similarly movies use 'persistence' of vision
  - 'decay time' of receptor response

Example: Plane propeller on phone camera

- http://www.youtube.com/watch?v=LVwmtwZLG88



Take Away

- After studying this material, you should be able to:
  1. Understand how discrete-time sampling works
  2. Understand the Shannon-Nyquist theorem
  3. Work out the Nyquist rate for signal of given bandwidth
  4. Work out the spectrum of a sampled signal given the original signal spectrum (and vice versa)
  5. Understand what aliasing is and how it can be handled
  6. Understand ideal and practical signal reconstruction methods